Question	Answer	Marks	Partial Marks
1	13 or – 13	**1**	
2	54	**2**	**M1** for $\frac{360}{8+5+4+3}[\times 3]$ or $\frac{3}{8+5+4+3}[\times 360]$ oe
3	11 27	**3**	**M1** for 500 ÷ 43 oe **M1** for $500 - \textit{their}\,11 \times 43$ oe *their* 11 must be an integer from 2 to 11
4	102	**1**	
5	180	**3**	**M2** for $[2\times](8\times 6+8\times 3+3\times 6)$ oe or **M1** for 8×6 or 8×3 or 3×6
6	0.48 oe	**2**	**M1** for $1-(0.2+0.32)$ oe
7	103.32 cao	**2**	**M1** for $126\times\left(1-\frac{18}{100}\right)$ oe or **B1** for 22.68
8	13 16 21	**2**	**B1** for 2 correct terms in correct position or **SC1** for 12, 13, 16
9	239	**2**	**M1** for 180 + 59 or 360 – (180 − 59) oe or indicates correct angle on diagram
10(a)(i)	$\begin{pmatrix}3\\4\end{pmatrix}$	**1**	
10(a)(ii)	$\begin{pmatrix}12\\48\end{pmatrix}$	**1**	
10(b)	5	**2**	**M1** for $(\textit{their}\,3)^2+(\textit{their}\,4)^2$ or better
11	24	**1**	
12(a)	correct graph	**3**	**B1** for line from (0, 0) to (1.5, 30) **B1** for horizontal line from (*their* 1.5, *their* 30) for 0.5 hours **B1** for a line from (*their* 2, *their* 30) ending at distance 70 with a gradient of 16 Provided it fits on the grid and *their* 30 is <70

Question	Answer	Marks	Partial Marks
12(b)	15.6 or 15.55 to 15.56 nfww	**3**	**M2** for 70 ÷ (*their* final time in hours) (final time =) $1.5 + 0.5 + \frac{70 - their\,30}{16}$ or 4.5 or *their* final time from graph or **M1** for 70 ÷ any time
13	$\frac{33}{8}$ or $\frac{17}{6}$ $\quad 2\frac{1}{8} - \frac{5}{6}$	**B1**	Correct step for dealing with mixed numbers Allow $\frac{33k}{8k}$ or $\frac{17k}{6k}$
	$\frac{99}{24}$ and $\frac{68}{24}$ $\quad [2]\frac{3}{24} - \frac{20}{24}$	**M1**	Correct method to find common denominator e.g. $4\frac{3}{24}$ and $2\frac{20}{24}$
	$1\frac{7}{24}$ cao and correct working	**A1**	
14	2.6[0] or 2.600…	**3**	**M2** for $\sqrt[10]{\frac{1328.54 + 4540}{4540}}$ or **M1** for $4540 \times k^{10} = 1328.54 + 4540$ for any k If 0 scored **SC1** for answer –11.6 or –11.56…
15	$4a^2b$ final answer	**2**	**M1** for two correct parts out of three from 4, a^2 and b in final answer
16(a)	$(M \cup G) \cap P'$	**1**	
16(b)	22	**1**	
16(c)	$\frac{8}{23}$ oe	**2**	**M1** for $\frac{k}{23}$ or $\frac{k}{3+9+5+6}$ or $\frac{8}{c}$ or $\frac{3+5}{c}$ $c \neq 1$ or for 8 and 23 identified
17(a)	 Correct sketch to go through (0, 0), (180, 0) and (360, 0)	**2**	**B1** for correct sine curve shape through the origin